

24-51-Q2

Q: m_1 m_2 C_1 C_2

(a) Bayes

(b) Naive Bayes

Solution (a) ① For a class w_i assumed to follow a multivariate normal distribution a suitable discriminant function is

$$g_i(x) = p(x|w_i) p(w_i)$$

$$= \frac{1}{(2\pi)^{\frac{d}{2}} |\hat{C}_i|^{\frac{1}{2}}} e^{-\frac{1}{2}(x_i - \hat{\mu}_i)^T \hat{C}_i^{-1} (x_i - \hat{\mu}_i)} p(w_i)$$

x is classified into class 1 if

$$g_1(x) > g_2(x)$$

x is classified into class 2 if

$$g_2(x) > g_1(x)$$

x is on the decision boundary (undecided)

$$\text{if } g_1(x) = g_2(x)$$

②

$$g_1(x) = \frac{1}{(2\pi)^{\frac{2}{2}} |\hat{C}_1|^{\frac{1}{2}}} e^{-\frac{1}{2}(x_1 - \hat{\mu}_1)^T \hat{C}_1^{-1} (x_1 - \hat{\mu}_1)} p(w_1)$$

$$= \frac{1}{2\pi \times \sqrt{1.02381}} e^{-\frac{1}{2} \times 1.3176} \times \frac{500}{500+300}$$

$$= 0.05087$$

$$\textcircled{3} g_2(x) = \frac{1}{(2\pi)^{\frac{2}{2}} |\hat{C}_2|^{\frac{1}{2}}} e^{-\frac{1}{2}(x_2 - \hat{\mu}_2)^T \hat{C}_2^{-1} (x_2 - \hat{\mu}_2)} p(w_2)$$

$$= \frac{1}{2\pi \times \sqrt{0.977049}} e^{-\frac{1}{2} \times 1.7118} \times \frac{300}{500+300}$$

$$= 0.02566$$

$$\textcircled{4} g_1(x_1) > g_2(x_2)$$

So the Bayes classifier assign x_1 to class 1

(b) ① Under the Naive Bayes assumption we treat each feature as independent. In effect, we replace the true covariance matrix C_i by its diagonal part

$$\sigma_{11}^2 = 0.942 \quad \sigma_{21}^2 = 1.087$$

$$\sigma_{12}^2 = 0.945 \quad \sigma_{22}^2 = 1.037$$

$$\mu_{11} = 0.026 \quad \mu_{21} = 0.048$$

$$\mu_{12} = 1.542 \quad \mu_{22} = 2.011$$

② Then, the Naive Bayes discriminant function for class w_i is the formula $g'_i(x)$.

$$g'_i(x) = \prod_{i=1}^n p(x_i | w_j) p(w_j)$$

$$p(x_i | w_j) = \frac{1}{\sqrt{2\pi} \sigma_{ij}} e^{-\frac{1}{2} \left(\frac{x_i - \mu_{ij}}{\sigma_{ij}} \right)^2}$$

x is classified into class 1 if
 $g'_1(x) > g'_2(x)$

x is classified into class 2 if
 $g'_2(x) > g'_1(x)$

x is on the decision boundary (undecided)
 if $g'_1(x) = g'_2(x)$

③ we denote $x_1 = 0.567$, $x_2 = 1.101$

$$p(x_1|w_1) = \frac{1}{\sqrt{2\pi} \sigma_{11}} e^{-\frac{1}{2} \left(\frac{x_1 - \mu_{11}}{\sigma_{11}} \right)^2}$$

$$= \frac{1}{\sqrt{2\pi} \times \sqrt{0.942}} e^{-\frac{1}{2} \left(\frac{0.567 - 0.026}{\sqrt{0.942}} \right)^2}$$

$$= 0.3501$$

$$p(x_2|w_1) = \frac{1}{\sqrt{2\pi} \times \sqrt{1.087}} e^{-\frac{1}{2} \left(\frac{1.101 - 0.048}{\sqrt{1.087}} \right)^2}$$

$$= 0.2298$$

$$p(x_1|w_2) = \frac{1}{\sqrt{2\pi} \times \sqrt{0.945}} e^{-\frac{1}{2} \left(\frac{0.567 - 1.542}{\sqrt{0.945}} \right)^2}$$

$$= 0.2482$$

$$p(x_2|w_2) = \frac{1}{\sqrt{2\pi} \times \sqrt{1.037}} \times e^{-\frac{1}{2} \left(\frac{1.101 - 2.011}{\sqrt{1.037}} \right)^2}$$

$$= 0.2628$$

$$\begin{aligned}
 g'_1(x) &= p(x_1 | w_1) p(x_2 | w_1) p(w_1) \\
 &= 0.3501 \times 0.2298 \times \frac{500}{800} \\
 &= 0.05028
 \end{aligned}$$

$$\begin{aligned}
 g'_2(x) &= p(x_1 | w_2) p(x_2 | w_2) p(w_2) \\
 &= 0.2482 \times 0.2628 \times \frac{300}{800} \\
 &= 0.02446
 \end{aligned}$$

$$g'_1(x) > g'_2(x)$$

So, Naive Bayes also classifies x_1 to Class 1